

SeRP Linkage Report

Splink Cohort Builder

Linkage Project Name: splink_cohort_builder

Linkage Run Name: Run 1

Linkage Type: Link Dedupe

Model Type: Probabilistic

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Run Configuration

Operation mode:	Link Dedupe
Linkage type:	Probabilistic
Fields used:	first_name, surname, dob, city, email, gender, postcode
Active blocking:	first_name, surname, dob, city, email, gender, postcode, first_name+surname
Cluster threshold:	0.8

Datasets

This section shows an overview of the datasets used, including linkage fields, field mapping, and column-level completeness (missingness).

Base table name: Dataset A (fake1000a)

Dataset: Splink fake_1000 + gender (name inference) + postcode (UK GeoNames)

Number of rows: 1000

Linkage fields

first_name, surname, dob, city, email, gender, postcode

Field mapping (Dataset A)

Linkage System Alias	Source Fieldname
first_name	first_name
surname	surname
dob	dob
city	city
email	email
gender	gender
postcode	postcode

Dataset statistics: Dataset A

Missingness: Completeness per column shown below.

Field	Completeness (%)
first_name	83.1%
surname	81.9%
dob	100.0%
city	81.3%
email	78.9%
gender	100.0%
postcode	57.7%

Base table name: Dataset B (fake1000b)

Dataset B is a 50% sample of Dataset A with controlled data quality errors: 14% first-name typos, 9% surname typos, 5% missing DOBs, 15% email variations, 11% city abbreviations, 7% gender errors.

Number of rows: 500

Field mapping (Dataset B)

Linkage System Alias	Source Fieldname
first_name	first_name
surname	surname
dob	dob
city	city
email	email
gender	gender
postcode	postcode

Dataset statistics: Dataset B

Field	Completeness (%)
first_name	83.2%
surname	82.6%
dob	94.2%
city	80.8%
email	77.4%
gender	100.0%
postcode	58.4%

Column Completeness Chart

Completeness = percentage of non-null values per column.

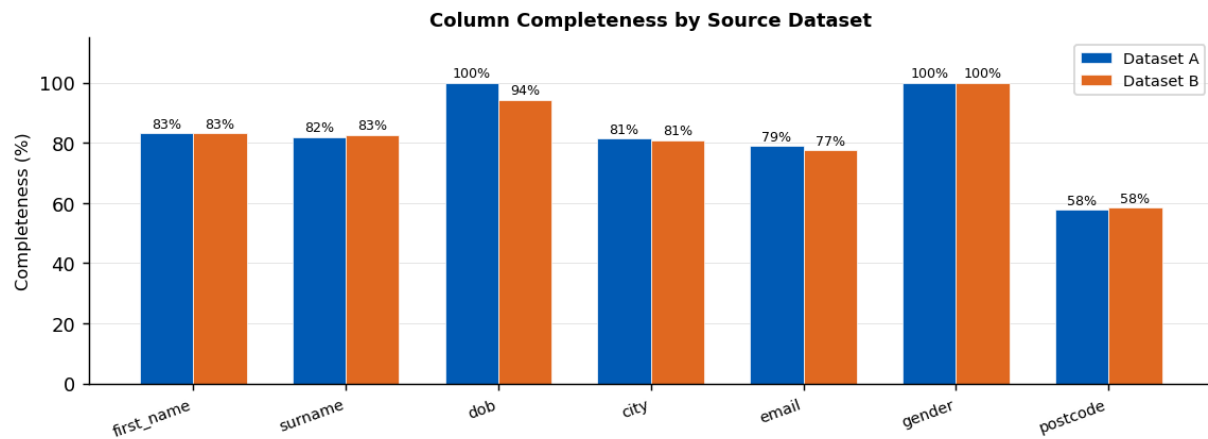


Figure: Column completeness by source dataset.

Blocking Rules

Blocking rules generate pairwise record comparisons. Only record pairs that agree on at least one blocking field are compared. Blocking reduces the $O(n^2)$ comparison space but may miss true matches if rules are too restrictive.

Blocking rule 0:

```
l."first_name" = r."first_name"
```

Number of generated comparisons: 1,003

Reasoning: Pairs that agree on this field are brought forward for detailed comparison. Useful when this field has high completeness and low error rate.

Blocking rule 1:

```
l."surname" = r."surname"
```

Number of generated comparisons: 540

Reasoning: Pairs that agree on this field are brought forward for detailed comparison. Useful when this field has high completeness and low error rate.

Blocking rule 2:

```
l."dob" = r."dob"
```

Number of generated comparisons: 339

Reasoning: Pairs that agree on this field are brought forward for detailed comparison. Useful when this field has high completeness and low error rate.

Blocking rule 3:

```
l."city" = r."city"
```

Number of generated comparisons: 288

Reasoning: Pairs that agree on this field are brought forward for detailed comparison. Useful when this field has high completeness and low error rate.

Blocking rule 4:

```
l."email" = r."email"
```

Number of generated comparisons: 75

Reasoning: Pairs that agree on this field are brought forward for detailed comparison. Useful when this field has high completeness and low error rate.

Blocking rule 5:

```
l."gender" = r."gender"
```

Number of generated comparisons: 280

Reasoning: Pairs that agree on this field are brought forward for detailed comparison. Useful when this field has high completeness and low error rate.

Blocking rule 6:

```
l."postcode" = r."postcode"
```

Number of generated comparisons: N/A

Reasoning: Pairs that agree on this field are brought forward for detailed comparison. Useful when this field has high completeness and low error rate.

Blocking rule 7:

```
l."first_name+surname" = r."first_name+surname"
```

Number of generated comparisons: N/A

Reasoning: Pairs that agree on this field are brought forward for detailed comparison. Useful when this field has high completeness and low error rate.

Cumulative count of blocking rules chart

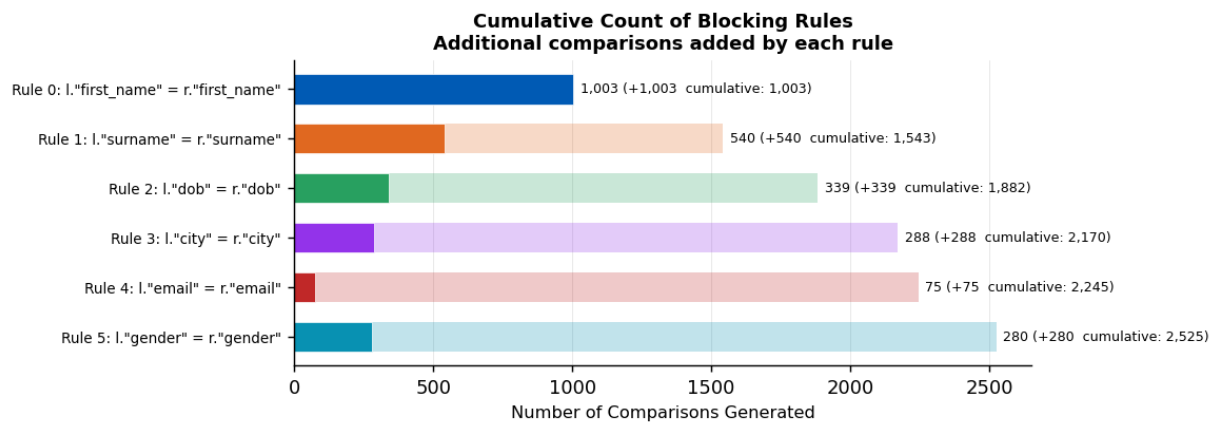


Figure: Additional comparisons generated by each successive blocking rule.

Comparison Methods

Comparison methods are used to score the relationship between two records. Each method defines ordered levels from exact match to total disagreement. A null level handles missing values with no evidence contribution.

Number of comparison methods for edge scoring:

Comparison 0: first_name

SQL comparison levels (first match wins):

```
"first_name_l" IS NULL OR "first_name_r" IS NULL [NULL LEVEL]
```

```
"first_name_l" = "first_name_r"
```

```
jaro_winkler_similarity("first_name_l", "first_name_r") >= 0.92
```

```
jaro_winkler_similarity("first_name_l", "first_name_r") >= 0.88
```

```
jaro_winkler_similarity("first_name_l", "first_name_r") >= 0.7
```

ELSE (all other comparisons)

Description: CustomComparison

Reasoning: Placeholder text.

Comparison 1: surname

SQL comparison levels (first match wins):

```
"surname_l" IS NULL OR "surname_r" IS NULL [NULL LEVEL]
```

```
"surname_l" = "surname_r"
```

```
jaro_winkler_similarity("surname_l", "surname_r") >= 0.92
```

```
jaro_winkler_similarity("surname_l", "surname_r") >= 0.88
```

```
jaro_winkler_similarity("surname_l", "surname_r") >= 0.7
```

ELSE (all other comparisons)

Description: CustomComparison

Reasoning: Placeholder text.

Comparison 2: dob

SQL comparison levels (first match wins):

```
try_strptime("dob_l", '%Y-%m-%d') IS NULL OR try_strptime("dob_r", '%Y-%m-%d') IS NULL [NULL LEVEL]
```

```
"dob_l" = "dob_r"
```

```
damerau_levenshtein("dob_l", "dob_r") <= 1
```

```
ABS(EPOCH(try_strptime("dob_l", '%Y-%m-%d')) - EPOCH(try_strptime("dob_r", '%Y-%m-%d')) <= 2629800.0
```

```
ABS(EPOCH(try_strptime("dob_l", '%Y-%m-%d')) - EPOCH(try_strptime("dob_r", '%Y-%m-%d')) <= 31557600.0
```

```
ABS(EPOCH(try_strptime("dob_l", '%Y-%m-%d')) - EPOCH(try_strptime("dob_r", '%Y-%m-%d')) <= 315576000.0
```

ELSE (all other comparisons)

Description: CustomComparison

Reasoning: Placeholder text.

Comparison 3: city

SQL comparison levels (first match wins):

```
"city_l" IS NULL OR "city_r" IS NULL [NULL LEVEL]
```

```
"city_l" = "city_r"
```

ELSE (all other comparisons)

Description: CustomComparison

Reasoning: Placeholder text.

Comparison 4: email

SQL comparison levels (first match wins):

```
"email_l" IS NULL OR "email_r" IS NULL [NULL LEVEL]
```

```
"email_l" = "email_r"
```

ELSE (all other comparisons)

Description: CustomComparison

Reasoning: Placeholder text.

Comparison 5: gender

SQL comparison levels (first match wins):

```
"gender_l" IS NULL OR "gender_r" IS NULL [NULL LEVEL]
```

```
"gender_l" = "gender_r"
```

ELSE (all other comparisons)

Description: CustomComparison

Reasoning: Placeholder text.

Comparison 6: postcode

SQL comparison levels (first match wins):

```
"postcode_l" IS NULL OR "postcode_r" IS NULL [NULL LEVEL]
```

```
"postcode_l" = "postcode_r"
```

ELSE (all other comparisons)

Description: CustomComparison

Reasoning: Placeholder text.

Model Training

This section shows the model training parameters and hyperparameters used for the probabilistic Fellegi-Sunter model.

Overview

Number of training schemes: 2 (prior estimate + EM)

Prior Estimation

Number of random samples: 1.00E+05

Recall estimate: 0.6

Rule 0 (prior estimation):

```
l."first_name" = r."first_name"
```

Estimation Schemes (Expectation-Maximisation)

fix_u_probabilities = True: u-probabilities held fixed during EM.

Training Rule 0:

```
l."first_name" = r."first_name"
```

Training Rule 1:

```
l."surname" = r."surname"
```

Unlinkable Records

Some records lack sufficient information to produce a match weight above a given threshold. This chart shows what percentage will be unlinkable at each threshold, guiding the choice of operating threshold.

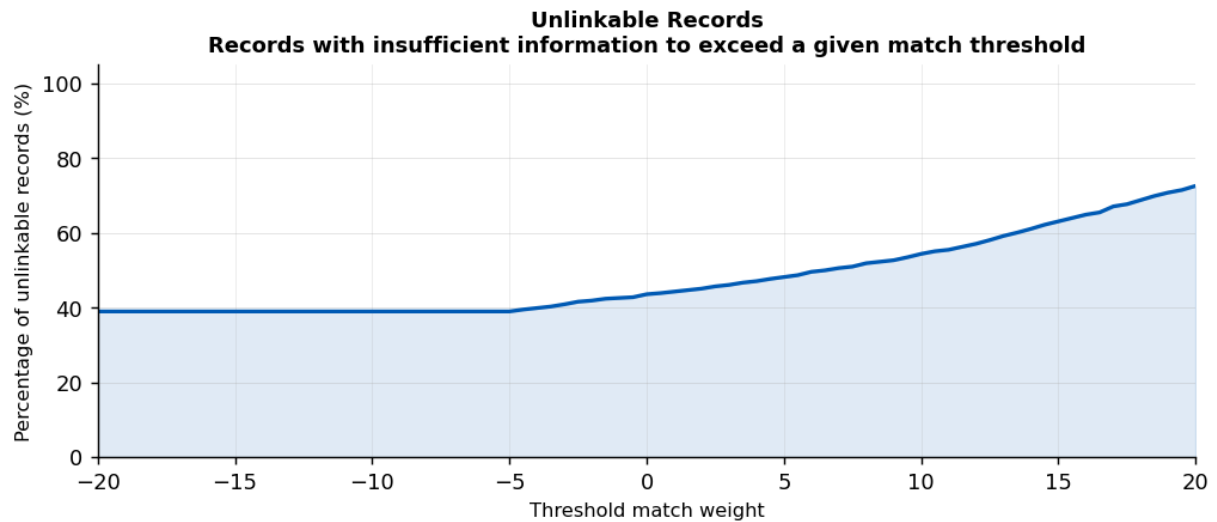


Figure: Percentage of unlinkable records vs match-weight threshold.

Edge Metrics

This section shows the results of model inference of pairwise edges. Quality review insights are provided for the number of edges predicted, distribution of match weights, and match probabilities.

Edge metrics overview:

Number of predicted edges: 2,525

Number of distinct unique_ids with an edge: 1,445

Match probability statistics

Statistic	Value
Mean	0.7355
Median	0.9929
Min	0.0307
Max	1.0
Std Dev	0.3792

Match Weight Histogram

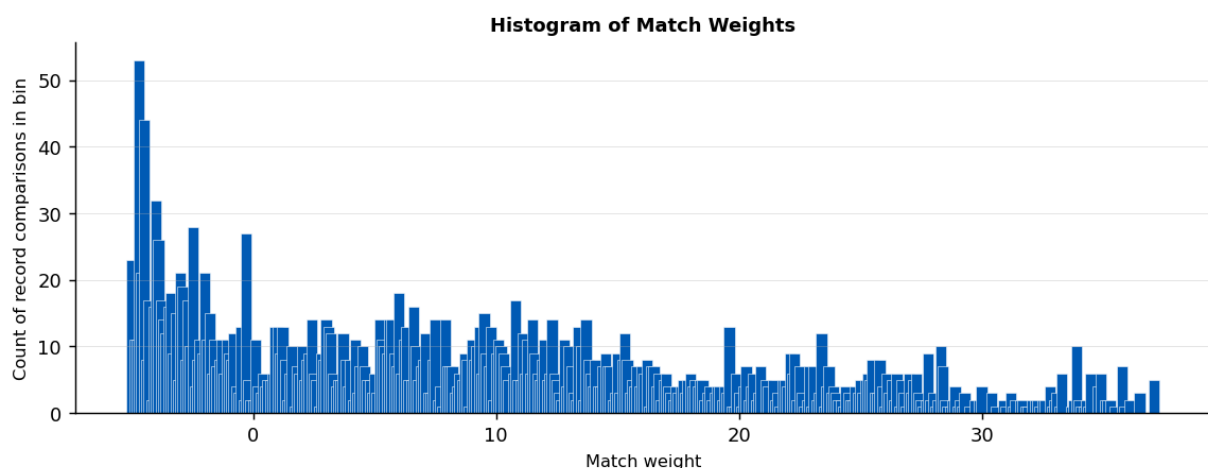


Figure: Histogram of match weights. Peaks near high positive values indicate confident predictions.

Mean gamma scores by field

Field	Mean Gamma
first_name	1.9204
surname	1.7564
dob	3.0214
city	0.1343
email	-0.0519
gender	0.7002
postcode	-0.3572

Cluster Metrics

This section shows the cluster inference results. A threshold was applied to the predicted edges to form entity clusters using a connected-components algorithm. Each cluster ideally represents one real-world individual.

Cluster metrics overview:

Number of clusters: 438

Cross-dataset clusters: 261

cluster_type	n_clusters	total_records
Multi-record cluster (2+ records)	261	1323.0
Singleton (1 record)	177	177.0

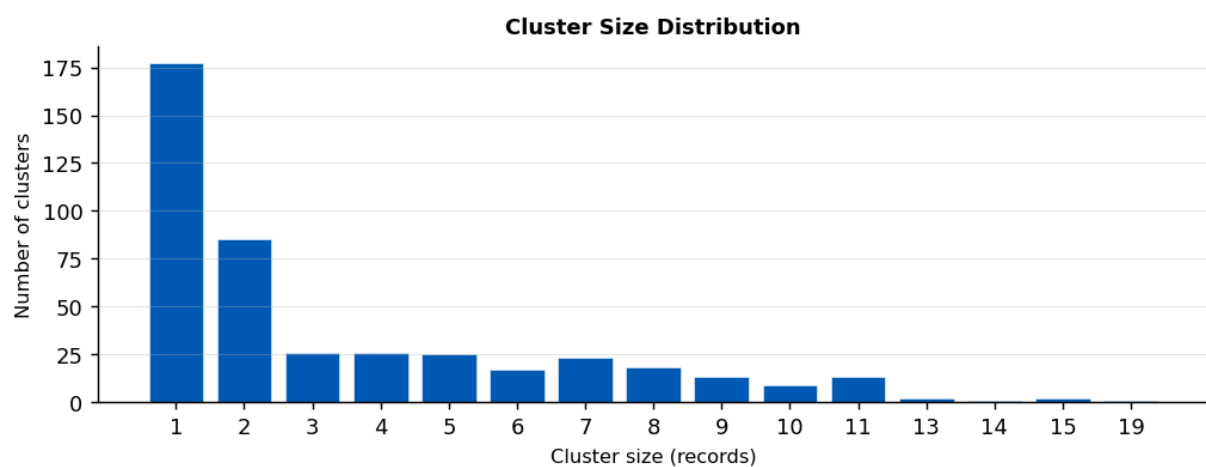


Figure: Cluster size distribution.

Dataset overlap Venn diagram

The Venn diagram shows how many clusters contain records from Dataset A only, Dataset B only, or both datasets. Cross-dataset clusters represent successfully linked records.

Category	N clusters
A only	176
B only	1
Both A and B	261

Venn Diagram of Cluster_id Set Membership fakea / fakeb

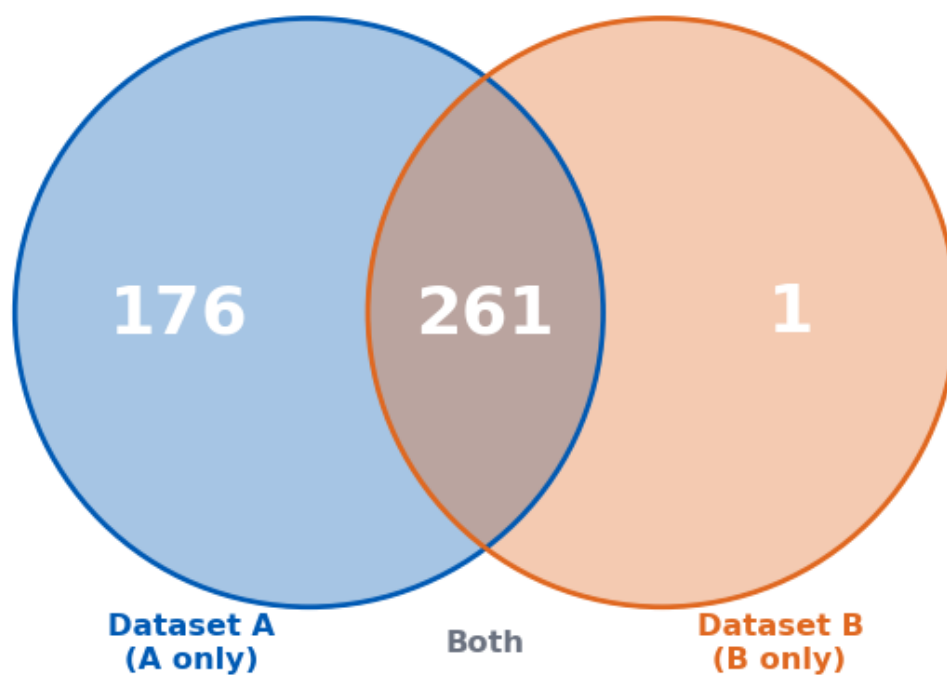


Figure: Venn diagram of cluster_id set membership.

Model Accuracy: Confusion Matrix

Ground truth: the 'cluster' column in the input datasets. Records with the same cluster value are true matches. TP = predicted match AND true match. FP = predicted match BUT not true match. FN = true match NOT predicted. TN is omitted (too large for pairwise comparison).

True Positives (TP):	2,030
False Positives (FP):	495
False Negatives (FN):	523
Ground truth pairs:	2,553
Predicted pairs:	2,525

Derived Metrics

Metric	Value	Interpretation
Precision	0.8040	$TP / (TP+FP)$ - of predicted matches, how many are correct
Recall	0.7951	$TP / (TP+FN)$ - of true matches, how many were found
F1 Score	0.7995	Harmonic mean of Precision and Recall
F* Score	0.6660	$TP / (TP+FP+FN)$ - accounts for both error types
FDR	0.1960	False Discovery Rate: $FP / (TP+FP)$
FNR	0.2049	False Negative Rate: $FN / (TP+FN)$

Confusion Matrix (Pairwise) Ground truth: 'cluster' column in original dataset

	Predicted Match	Predicted Non-Match
True Match	TP 2,030	FP 495
True Non-Match	FN 523	TN (omitted)

Precision: 0.8040 Recall: 0.7951 F1: 0.7995
F*: 0.6660 FDR: 0.1960 FNR: 0.2049

Figure: Confusion matrix (pairwise). TN omitted as it is too large to count practically.

Precision-Recall Curve and F* Score (example 16 - CRL analysis)

These curves show Precision and Recall at every match-probability threshold. The F* curve shows the composite score $TP/(TP+FP+FN)$. The ideal operating point maximises both Precision and Recall.

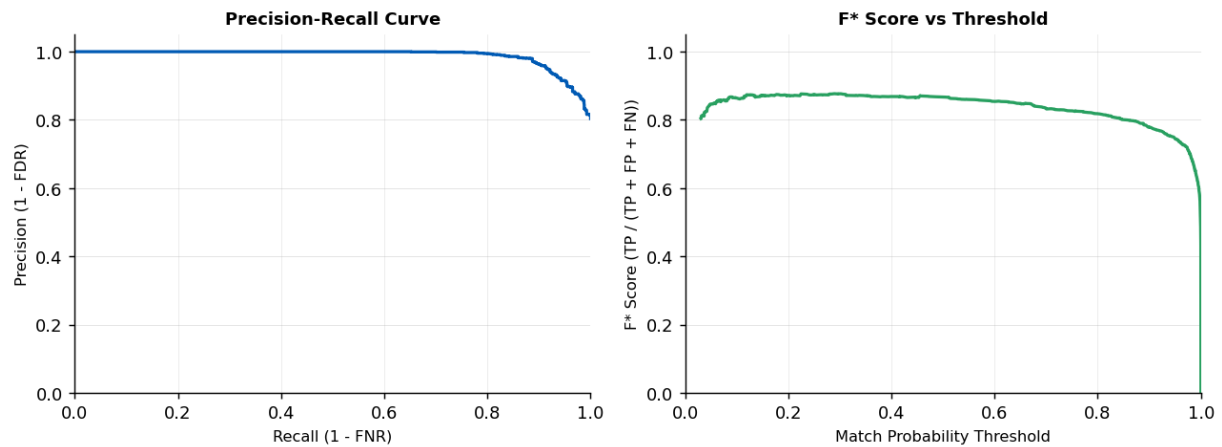


Figure: Precision-Recall curve (left) and F* vs threshold (right).

CRL Score (Composite Reliability of Linkage)

The CRL score measures linkage quality in the valid operating region where both FDR (False Discovery Rate) and FNR (False Negative Rate) are within the acceptable epsilon threshold. $CRL = AVG(F^*) * (t_{upper} - t_{lower})$. A higher CRL score indicates more reliable linkage.

Epsilon (tolerance):	0.10
CRL Score:	0.215248
t_upper:	0.35720637278584455
t_lower:	0.11043126249640364
epsilon_z:	0.07070707070707072